SUNY Old Westbury 2026-2027 Undergraduate Catalog

Mathematics, B.S.

HEGIS Number: 1701

Curriculum Code: U0334

Mathematics encompasses logic and reasoning skills that provide the tools for making decisions, interpreting observations, explaining natural phenomena, and solving problems. The Department aims to instill in students an appreciation of mathematics, and to equip students with the skills needed to achieve academic and career goals within a complex technological society. In addition to serving mathematics majors, the Department offers for business, science and liberal arts majors a number of courses that concentrate on problem solving.

The mission of the Bachelor of Science Degree in Mathematics is to teach students the fundamentals in a broad range of areas of mathematics, develop their skills in higher reasoning, logic, proof, abstraction, and mathematical modelling, and enable them to communicate mathematics effectively. The program prepares students for careers in the corporate, government and education sectors as well as for graduate school.

Mission of the Bachelor of Science Degree in Mathematics

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Learning Outcomes

1. Mathematics Fundamentals. Demonstrate basic mathematical understanding and computational skills. Calculus, Linear Algebra, Differential Equations
2. Effective Reasoning and Communication. Effectively communicate mathematical reasoning in a precise and articulate manner. Transition to advanced mathematics
3. Advanced Mathematics. Demonstrate the ability to make rigorous arguments in advanced mathematical topics: Abstract Algebra and Advanced Calculus

Career Opportunities

The study of undergraduate Mathematics provides essential preparation for a variety of professional careers. Throughout government and industry, mathematical modeling and quantitative methods lie at the heart of research, analysis and problem solving. Old Westbury's Mathematics graduates have pursued graduate studies and successful careers in fields such as actuarial science, data science, management science, finance, industrial engineering, operations research, and statistics.

Mathematics Learning Center

The Mathematics Learning Center (MLC) provides academic support for students taking mathematics courses and encourages students to develop a deeper appreciation and interest in mathematics. Staffed by professional educators and peer tutors, the MLC welcomes all students. The telephone number is 516-628-5622 Specific hours are announced each semester.

Requirements for a Bachelor of Science (B.S.) Degree in Mathematics

Mathematics Major Requirements: 56 Credits (14 courses)

1. Required Core Courses: 40 Credits (10 courses)

- MA 2310 - Calculus & Analytic Geometry I Credits: 4
- MA 2320 - Calculus & Analytic Geometry II Credits: 4
- MA 3030 - Discrete Mathematics Credits: 4
- MA 3160 - Linear Algebra Credits: 4
- MA 3210 - Introduction to Probability and Statistics Credits: 4
- MA 3330 - Calculus and Analytic Geometry III Credits: 4
- MA 3520 - Transition to Advanced Mathematics Credits: 4
- MA 4360 - Differential Equations Credits: 4
- MA 5120 - Abstract Algebra I Credits: 4
- MA 5320 - Advanced Calculus I Credits: 4

2. Computer Programming Requirement: 4 Credits

(1 course)

- CS 2510 - Computer Programming I Credits: 4

  OR

- CS 2521 - Scientific Programming in Python Credits: 4

3. required Elective Courses: 12 Credits (3 Courses)

- Any 4000 or 5000 level Mathematics Course (Each 4 Credits)

- CS 3810 - Data Structures and Algorithms Credits: 4

Liberal Education Curriculum Courses

Refer to the Liberal Education Curriculum Bulletin for requirements.

General Electives

In consultation with academic advisor, for a total of 120 credits.

Other Requirements

- A grade of **C** or higher is needed in all required mathematics and computer science courses.
- Transfer students must complete a minimum of 28 credits (7 courses) of the required major courses at or above the 3000 level at Old Westbury.

College Wide Requirement

- 120 credits overall (40 credits at Old Westbury, may transfer a maximum of 80 credits)
- 45 Upper Division credits (3000, 4000, or 5000 level courses)
- 60 Liberal Arts credits
- Cumulative Grade Point Average of 2.0

Second Major in Adolescence Education Mathematics

Candidates who desire a second major in Adolescence Education Mathematics must complete the requirements for Adolescence Education Mathematics and take: MA 4360, CS 2510, and one Math elective at 4000 or 5000 level in addition to MA 4510, MA 5120, MA 5320. You must declare Mathematics as the first major and Adolescence Education Mathematics as the second.

Sample Course Sequence By Major

Term 1

- MA 2090 - Precalculus Credits: 4

Term 2

- MA 3030 - Discrete Mathematics Credits: 4
- MA 2310 - Calculus & Analytic Geometry I Credits: 4
- CS 2510 - Computer Programming I Credits: 4

Term 3

- MA 2320 - Calculus & Analytic Geometry II Credits: 4
- MA 3160 - Linear Algebra Credits: 4
- MA 3210 - Introduction to Probability and Statistics Credits: 4

Term 4

- MA 3330 - Calculus and Analytic Geometry III Credits: 4
- MA 3520 - Transition to Advanced Mathematics Credits: 4

Term 5

- MA 5120 - Abstract Algebra I Credits: 4
- MA 4360 - Differential Equations Credits: 4

Term 6

- MA 5320 - Advanced Calculus I Credits: 4
- Elective

Term 7

- Elective

Term 8

- Elective